

# CYBERSECURITY – STUDENT STUDY MATERIAL

A short beginner-friendly revision guide covering the most important cybersecurity concepts.

## 1. What is Cybersecurity?

**Cybersecurity** is the practice of protecting computers, networks, applications, devices and data from unauthorized access, misuse, damage or disruption. It is a continuous process because threats and technologies keep changing.

## 2. CIA Triad – Three Core Principles

| Principle       | Meaning                                                                | Example                    |
|-----------------|------------------------------------------------------------------------|----------------------------|
| Confidentiality | Only authorized people can access information.                         | Passwords, access controls |
| Integrity       | Information remains accurate and is not changed without permission.    | File protection, hashing   |
| Availability    | Systems and information are available when authorized users need them. | Backups, reliable systems  |

## 3. Common Cyber Threats

**Malware:** Malicious software designed to compromise devices or data. Examples include viruses, worms, Trojans and spyware.

**Phishing:** Fraudulent messages that try to trick users into clicking harmful links, opening files or revealing sensitive information.

**Ransomware:** Malware that can block access to data or systems and demand payment from victims.

**Social Engineering:** Manipulating people into giving information or performing an action that helps an attacker.

**Password Attacks:** Attempts to obtain or guess passwords, or to reuse passwords exposed in previous breaches.

**Identity Theft:** Using another person's personal information without permission.

## 4. Authentication and Access Control

**Authentication** verifies who a user is. **Authorization** determines what that user is allowed to access. **Multi-factor authentication (MFA)** adds another verification factor, such as an authenticator app, security key or code.

## 5. Cryptography Basics

**Encryption** converts readable information (plaintext) into protected ciphertext so unauthorized people cannot easily read it. **Decryption** converts it back when authorized. **Hashing** produces a fixed-length value from data and is commonly used to protect stored passwords. **Digital signatures** help verify authenticity and integrity.

## 6. Network Security

**Firewall:** Controls network traffic according to security rules.

**VPN:** Creates a protected connection over a network.

**Network segmentation:** Divides a network into smaller sections to limit the spread or impact of attacks.

**HTTPS:** Uses encryption to protect communication between a browser and a website.

## 7. Cyber Hygiene – Important Safety Rules

1. Use long, unique passwords or passphrases for different accounts.
2. Enable MFA whenever it is available.
3. Do not click unexpected links or open suspicious attachments.
4. Verify urgent requests using trusted contact information.
5. Keep operating systems, browsers and applications updated.
6. Use reputable antivirus/anti-malware protection where appropriate.
7. Back up important files regularly and protect the backups.
8. Do not share passwords, OTPs, PINs or recovery codes with others.
9. Use secure networks and avoid entering sensitive information on untrusted Wi-Fi.
10. Lock your device when you are away from it.

## 8. What to Do After a Suspected Phishing Attack

Stop interacting with the suspicious message. If credentials were entered, change the affected password immediately and change it anywhere else it was reused. Enable MFA, report the incident to the relevant service or organization, and monitor the affected account for unusual activity.

## 9. Important Cybersecurity Terms

**Vulnerability:** A weakness that can be exploited.

**Threat:** A potential cause of harm to a system or data.

**Risk:** The possibility and potential impact of a harmful event.

**Exploit:** A technique or code that takes advantage of a vulnerability.

**Patch:** A software update that fixes bugs or security weaknesses.

**Backup:** A separate copy of data used for recovery.

**Incident:** A security event that may compromise confidentiality, integrity or availability.

## 10. Quick Revision Questions

1. What is cybersecurity?
2. What are the three parts of the CIA Triad?
3. What is phishing? Give two warning signs.
4. What is malware? Name two types.
5. What is the difference between authentication and authorization?
6. Why is MFA useful?
7. What is encryption?
8. Why should software be updated regularly?
9. What is a firewall?
10. Write five cybersecurity safety rules for students.

Study note: This material is intended for beginner-level revision. It is based on introductory cybersecurity guidance and educational resources from NIST and other academic sources.